

COMPUTER SCIENCE • CLASS 10

Introduction to Operating Systems

A comprehensive academic guide exploring system software, fundamental device operations, user interface models, desktop Titans, and hardware resource coordination.

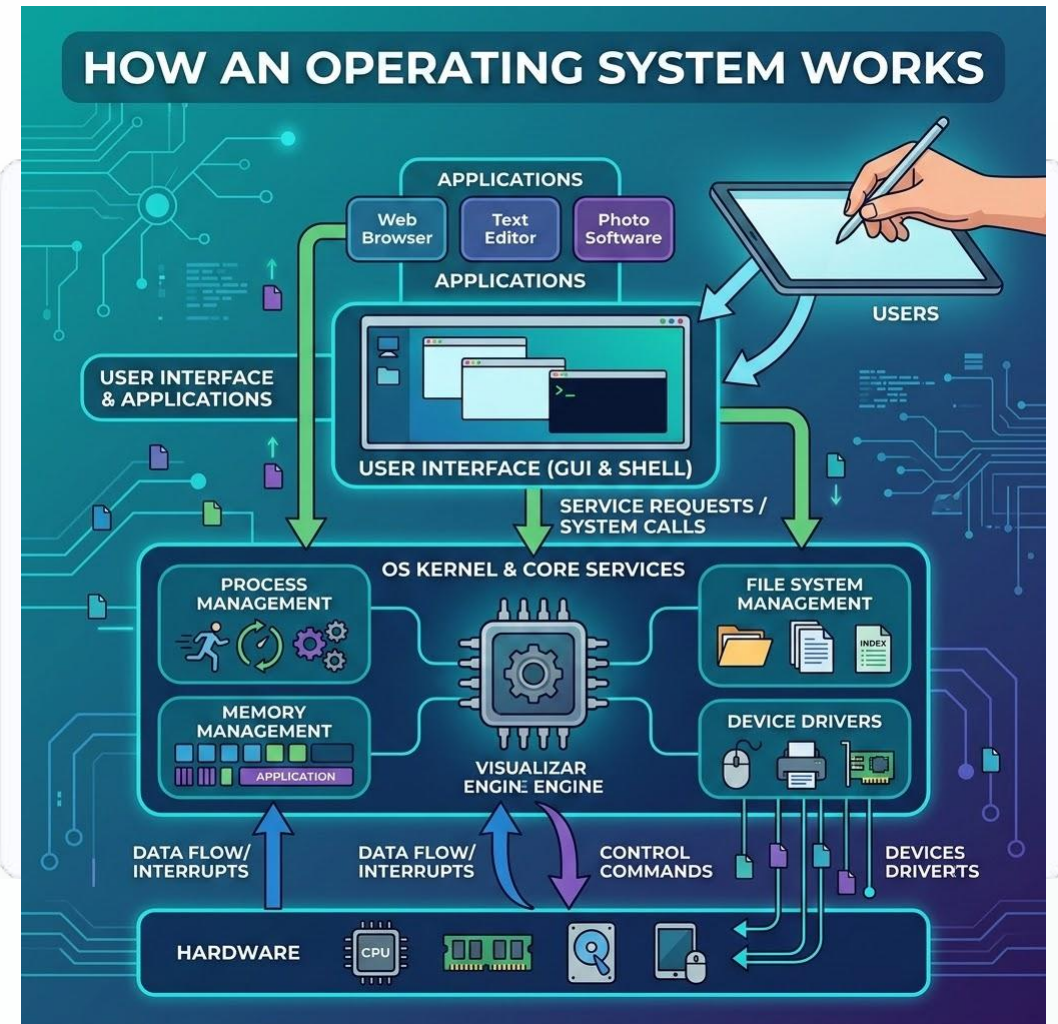
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What is an Operating System?

The Gateway System Software

An **Operating System (OS)** is a collection of programs that controls and coordinates all the tasks and functions in a computer system. It mediates between application programs and hardware components.

- ◆ **Hardware Intermediary:** Functions as a translator/interface directly between raw electronic computer hardware and human users.
- ◆ **Hardware Efficiency:** Empowers standard computer hardware systems to execute processing tasks with maximum efficiency and minimum delay.
- ◆ **Unified Program Coordinator:** Coordinates, boots, schedules, and isolates secondary application processes to avoid software crashes.



| Primary Functions of an OS



Process Management

A Process management is an application or program in running state. Process management is the important part of an operating system which enables the **activities** of planning, monitoring, and performance of a CPU.



Process Management

As a memory manager, the os handles the allocation and deallocation of memory space as required by various programs. Managing the primary memory, sharing and minimising memory access time are the basic goals of the memory management.



File Management

The operating system is responsible for creation and deletion of files and directories. It also takes care of other file related activities such as organising, storing, retrieving, naming and protecting the files.



Input/Output Management

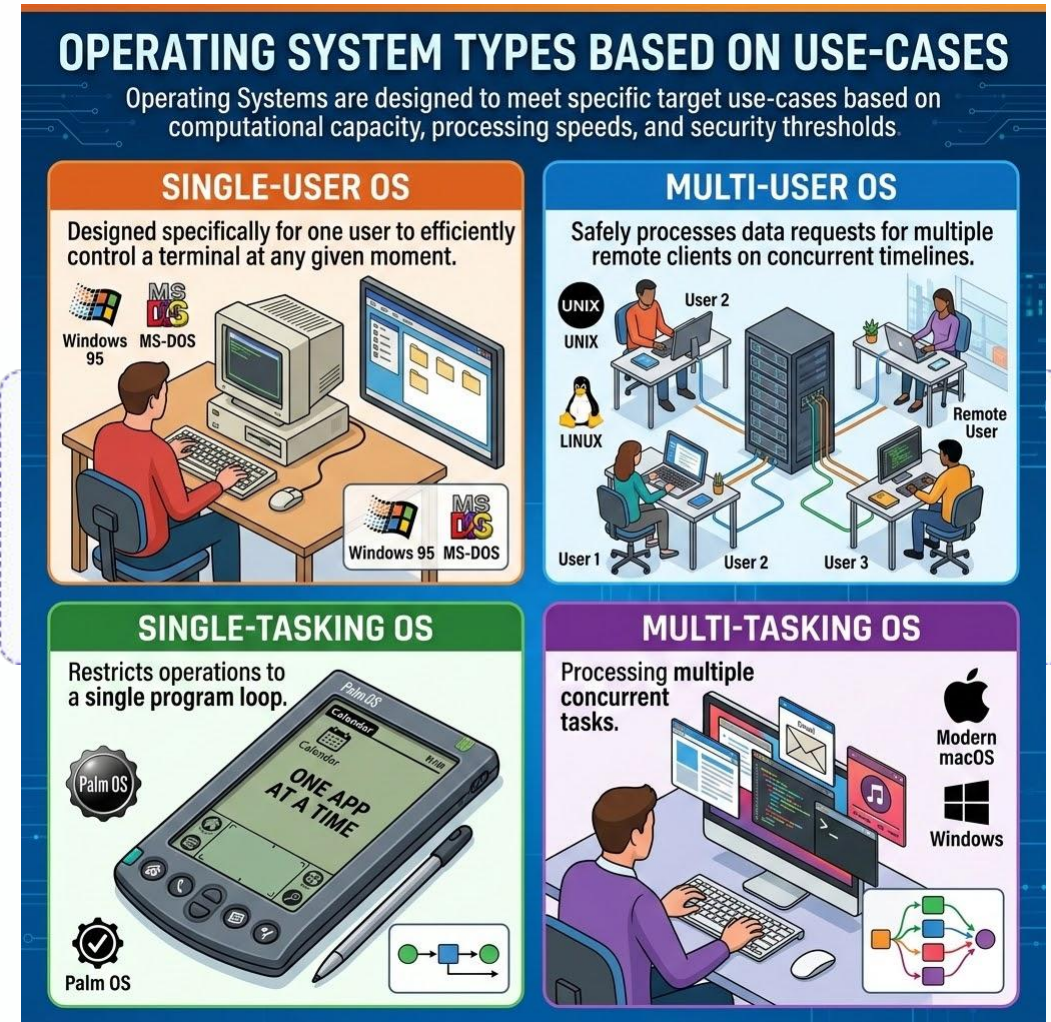
The input/output management module of an operating system coordinates and assigns different input and output devices. It controls all input/output devices, keeps track of input/output requests, issues commands to these devices.

Core Operating System Classifications

Understanding Structural Differences

Operating Systems are designed to meet specific target use-cases based on computational capacity, processing speeds, and security thresholds:

- ♦ **Single-User OS:** Designed specifically for one user to efficiently control a terminal at any given moment (e.g., Windows 95, MS-DOS).
- ♦ **Multi-User OS:** Safely processes data requests for multiple remote clients on concurrent timelines (e.g., UNIX, Linux).
- ♦ **Single-Tasking vs Multi-Tasking:** Restricts operations to a single program loop (Palm OS) versus processing multiple concurrent tasks (Modern macOS / Windows).



Operating System Classifications Matrix

OS Classification Type	Core Functional Feature Description	Standard Platform Example
Time Sharing	Rapid processor interval allocation allowing virtual execution concurrency	Modern Apple macOS
Real-Time (RTOS)	Ultra-strict deterministic executions with zero delay tolerances	VxWorks / QNX / RTLinux
Distributed System	Splits computational tasks across independent physical server nodes	LOCUS Architecture
Batch-Processing	Executes pre-grouped lists of automated jobs in a strictly linear format	IBM OS/2 / Early Mainframes
Multi-Processing	Orchestrates tasks utilizing two or more physical computer CPUs simultaneously	Unix / Linux / Windows 2000

| Time Sharing Operating System

In this OS, a small amount of time is allocated for the processing simultaneously. It provides scheduling to each process to be run on.

| Real-Time Operating System

It was designed to respond to an event within a predetermined time. It aims to execute real time applications such as auto pilot mechanism used in aircrafts. It has very little user-interface capability and no end-user utilities.

| Single-User Operating System

It is designed for one user to effectively use a computer at a time.

| Multi-User Operating System

It allows many users to take advantage of the computer's resources simultaneously. It ensures that problems with one user doesn't affect the entire community of users.

| Batch-Processing Operating System

Similar types of jobs are grouped under batch and these batches are executed without user's interruption. Jobs are processed in the order of submission.

| Desktop Systems: UNIX & Linux

UNIX (First Released 1969)

An influential standard system codebase configured originally inside legendary Bell Laboratories.

- ♦ **Written in C:** Enabled easy execution portability to other CPU instruction architectures.
- ♦ **Pioneered Multitasking:** Allows clean isolated runtime frames protecting computer memory pools.

Linux (Open Source Alternative)

A famous modular open-source operating kernel created for standard personal desktop PCs and massive servers.

- ♦ **Unrivaled Security:** Exceptionally resilient layout constructed with native virus-free protections.



Portability Breakthrough

Because UNIX was written in the highly standard **C programming language** rather than complex machine-level Assembly language, engineers could instantly compile it for any computer hardware processor.

The Evolution of Microsoft Windows

01

Win NT

1993

New Technology standard. Multi-user focus.

02

Win 95/98

1995 - 1998

Consumer oriented, officially removed DOS reliance.

03

Win XP

2001

Standardized user UI, high system stability.

04

Win 7/8

2009 - 2012

Widely used UI, transition to mobile tablet UI views.

05

Win 10/11

2015 - Pres.

Modern fast kernel, cloud services integration.

| Alternative Desktop Environments



Apple Mac OS

Introduced in March 2001. Built on a powerful, secure UNIX kernel foundation. Known for stability, elegant interface, and exclusive hardware integration limits.



Oracle Solaris

Developed originally by Sun Microsystems in 1993. Scalable, high-security networking platform targeting heavy enterprise databases and enterprise cloud networks.



MS-DOS (Legacy)

16-bit Disk Operating System released in 1981 for Intel 8086 CPUs. Command-line interface operating from floppy disks with extremely tiny memory footprints.

| Mobile Operating Systems

Pocket-Sized Software Kernels

Smartphone operating systems are specifically configured to coordinate touch displays, wireless cellular signals, and critical battery-saving cycles.

- ♦ **Google Android:** Linux-based open-source platform launched in September 2008. Designed to support multi-vendor touch screen phones.
- ♦ **Apple iOS:** Highly optimized closed-source operating ecosystem designed in June 2007 exclusively for Apple iPhones.
- ♦ **BlackBerry & Symbian:** Legacy systems (1997–1999) that pioneered mobile messaging but were replaced in the touch-first era.

📋 Modern Focus Checklist

Android: Highly customizable, uses Android Runtime (ART).

iOS: Tight security models, extreme graphics framework optimization.

HCI Interface: CUI vs GUI



Character User Interface (CUI)

Users interact with the system only by typing alphanumeric instruction phrases into terminal prompts.

```
C:\> dir /s /w /p
```

Requires learning complex system commands by heart. Almost zero graphical processing demands on system hardware.



Graphical User Interface (GUI)

Users interact by visually clicking graphic indicators using mouse cursors or finger-touch monitors.

Folder Icon

Window

Highly intuitive, zero programming knowledge required. Demands active graphic processors and larger system RAM allocations.

| Innovation: AI-Native Operating Systems

The Future of System Software

Class 10 students should understand where computers are heading.

Standard OS architectures are shifting towards **AI Integration**:

- ◆ **AI-Driven Process Scheduling:** Neural-engine prediction maps predict app open workflows, optimizing system battery savings.
- ◆ **Voice & Agent-First UI Shells:** Natural language voice prompts are replacing traditional CLI or icon clicking paradigms.
- ◆ **Local Co-Processing:** Splitting operations cleanly between local NPU chips and secure system memory limits.

Tomorrow's Career Prep

Within the next decade, standard folders might be replaced by "semantic search streams" where the OS naturally understands what document you want simply from context clues.

Computer Care & Maintenance Checklist



Disk Cleanup

Run regular system sweeps to remove old cached system data and duplicate folders, keeping the storage run path clean.



System Updates

Install critical security updates regularly to protect against potential internet threat backdoors.



Dust Prevention

Physically blow out computer fans at regular intervals to prevent temperature throttling on the motherboard.



Malware Defense

Keep trusted antivirus software definitions active to screen downloads from malicious file injection attempts.

 **Practice Lab: Students will perform file defragmentation and disk cleanup tasks in tomorrow's laboratory!**